M13D — Mock exam 5

Instruments · Rotary Wing Aerodynamics · Autoflight · Equipment and Furnishings

40 questions · 20 Instruments · 13 Autoflight · 5 Rotary Wing Aerodynamics · 2 Equipment and Furnishings

Level 3: 22 · Level 2: 13 · Level 1: 5. Three options, one correct.

Every question sits on a fact highlighted or underlined in the manual. No question in this paper appears in any of the other four.

The answer key follows the questions, with the explanation and the manual PDF page for each one.

INSTRUMENTS

1. A pilot reads his magnetic heading and a VOR course deviation from one round instrument. Which indicator is he using?

- **A.** The HSI
- **B.** The ADI
- **C.** The variometer

2. How does the temperature behave with increasing height in the troposphere

- **A.** It decreases by 6.5 °C every 100 metres
- **B.** It increases by 0.65 °C every 100 metres
- **C.** It decreases by 0.65 °C every 100 metres

3. The indications fluctuate during a check of the pitot static system. What does this show?

- **A.** The presence of water in the pressure pipes
- **B.** The presence of ice at the static port
- **C.** The presence of a leak in the pressure pipes

4. A new reference pressure is selected on an electrical altitude indicator. How is the correction made?

- **A.** Pneumatically, by an additional set of bellows
- **B.** Mechanically, by turning the whole altimeter mechanism
- **C.** Electrically, in the instrument or in an air data computer

5. Why can the speed limit of a jet aeroplane at low altitude be read directly as an indicated airspeed?

- **A.** Because the limit there depends on the speed of sound, and the airspeed indicator measures dynamic pressure
- **B.** Because the limit there depends on dynamic pressure, and the airspeed indicator measures total pressure
- **C.** Because the limit there depends on dynamic pressure, and the airspeed indicator measures dynamic pressure

6. A machmeter is opened and found to contain a differential pressure capsule and an aneroid capsule. Which instruments use these same two sensing elements?

- **A.** The altimeter uses a differential pressure capsule and the airspeed indicator an aneroid capsule
- **B.** The airspeed indicator uses a differential pressure capsule and the altimeter an aneroid capsule
- **C.** The airspeed indicator uses a differential pressure capsule and the vertical speed indicator an aneroid capsule

7. The data supplied by the air data components of a GNADIRU comprises

- **A.** attitude, heading, vertical acceleration and body rates
- **B.** airspeed, Mach, angle of attack, temperature and barometric altitude
- **C.** present position, satellite status and receiver clock error

8. An altitude deviation occurs in the cruise. What does the crew get on EICAS and aurally?
   - A. The warning message ALTITUDE ALERT with Master Warning and a bell
   - B. An advisory message ALTITUDE ALERT with no aural indication
   - C. The caution message ALTITUDE ALERT with Master Caution and the owl tone

9. What does a running gyroscope tend to do
   - A. align its axis with the earth's magnetic field
   - B. maintain the direction of its axis in space
   - C. hold its axis square to the aircraft structure

10. What is the horizontal component of apparent wander called
    - A. apparent drift
    - B. transport wander
    - C. apparent topple

11. A gyro has worn frame bearings and is also carried east-west across the earth. How do the two error sources add up?
    - A. the bearing error takes the place of the earth rotation and transport errors
    - B. the bearing error sits on top of the earth rotation and transport errors
    - C. the bearing error is cancelled by the earth rotation and transport errors

12. A gyro on the bench has its rotational axis standing upright. Which measuring task is it built for?
    - A. turn rate measurement
    - B. attitude measurement
    - C. directional measurement

13. The gyro axis points at the centre of the earth during unaccelerated flight. What are the two erection balls doing?
    - A. they are balanced out, so no momentum acts on the gyro axis
    - B. they are held at their stops by the gyro cut-out rocker
    - C. they are driven apart, so a momentum acts on the gyro axis

14. The roll level switch of an electrically driven gyro detects the axis in an incorrect direction. What follows?
    - A. the roll torque motor applies a force and precession corrects it
    - B. the roll cutout switch opens and the erection circuit is broken
    - C. the caging mechanism locks the gimbals in a normal position

15. A rate of turn gyro is held by a linear spring. What can then be said about its precession angle?
    - A. it is proportional to the angular velocity about the measuring axis
    - B. it falls as the angular velocity about the measuring axis rises
    - C. it is independent of the angular velocity about the measuring axis

16. The system has been in DG mode for a long time and the heading has slowly wandered. What must the crew do?
    - A. correct the indication with the SET HDG knob
    - B. switch briefly to the slaved mode and straight back again
    - C. wait for the synchronising annunciator to null it out

17. An HSI shows several indications driven from different sources. Which one of them is driven by the heading system?
    - A. the compass rose
    - B. the course bar
    - C. the distance readout

18. The approaching minimums option is selected with the aural approaching minimums. How far above the decision height does the call-out normally come on?
    A. 100 feet
    B. 80 feet
    C. 130 feet

19. Which alerting envelopes does the terrain alerting function compute
    A. A take-off envelope and a final approach envelope
    B. A look-up envelope and a lateral offset envelope
    C. A caution-level envelope and a warning-level envelope

20. The terrain display relays have been energised by the alert discretes. What do the navigational displays then carry?
    A. The weather radar picture as before
    B. The conflicting ground in yellow or red
    C. The terrain test pattern in cyan

AUTOFLIGHT

21. In an I-regulator the rate of change of the output value is proportional to
    A. its input value
    B. the rate of change of its input value
    C. the change in its output value

22. With one exception, direct inputs affecting the translatory degrees of freedom are not possible. How then is the flight path controlled?
    A. Indirectly, through inputs affecting initially the rotary degrees of freedom
    B. Directly, through inputs applied to the three translatory degrees of freedom
    C. Indirectly, through inputs affecting initially the centre of gravity position

23. Which parts of the autoflight system provide the combinations of regulators used for automatic vertical flight path guidance?
    A. The AP/FD pitch channel and the flight management system under ATC control
    B. The AP/FD pitch channel and the auto-thrust system under AP/FD control
    C. The AP/FD roll channel and the auto-thrust system under FMCS control

24. Why are flight condition regulators called basic regulators?
    A. The inner control loop uses them to achieve attitude control through flight path control
    B. The outer control loops use them to achieve flight path control through attitude control
    C. The outer control loops replace them whenever flight path control is required

25. The autopilot is disengaged after a long cruise during which auto-trim has been working. Why is there no abrupt pitch attitude change?
    A. The elevator is already held by the torque limiter
    B. The stabilizer is already resting in its neutral position
    C. The elevator is already resting in its neutral position

26. The autopilot has just been engaged and no upper mode has been selected. Which attitudes do its two channels control straight away?
    A. The pitch channel the pitch attitude, the roll channel the roll attitude or heading
    B. The pitch channel the pitch attitude, the roll channel the heading only
    C. The pitch channel the flight path, the roll channel the roll attitude or heading

27. The crew has a heading mode running and air traffic control passes a different heading. When may the selected heading be altered?

- **A.** Only after the mode has been reselected
- **B.** Only after the first heading has been captured
- **C.** At any time while the mode is active

28. Why does the autopilot have to treat interception and stabilization as two separate jobs?

- **A.** The path is continuous, but the sensors update at two different rates
- **B.** The path is in segments, and each must be caught without overshoot or offset
- **C.** The path is in segments, and each one has to be flown at a different airspeed

29. The level off mode has just been selected during a climb. What does the vertical speed controller do?

- **A.** It reduces the actual rate by half and then holds it
- **B.** It holds the altitude reached when the mode was selected
- **C.** It reduces the actual rate to zero and holds zero

30. The landing controller adds a height bias to the measured radio altitude close to the ground — what does this achieve at the wheels?

- **A.** A sink rate is still commanded once the aircraft is on the ground
- **B.** A thrust increase is still commanded once the aircraft is on the ground
- **C.** A pitch up is still commanded once the aircraft is on the ground

31. During an approach the angle of attack becomes extreme. Which protection acts and with which thrust?

- **A.** Alpha floor protection, with go around thrust
- **B.** Overspeed protection, with idle thrust
- **C.** Low speed protection, with climb thrust

32. An operator intends to fly in all weather conditions — which flight rules must its approval cover?

- **A.** Visual flight rules
- **B.** Instrument flight rules
- **C.** Special visual flight rules

33. The aircraft reaches the decision height on a CAT II approach — what applies to a manual landing?

- **A.** It is forbidden in every case
- **B.** It is compulsory in every single case
- **C.** It is allowed but is not compulsory

ROTARY WING AERODYNAMICS

34. What is the advancing blade

- **A.** The forward moving blade
- **B.** The blade at the tail boom
- **C.** The backward moving blade

35. Why does raising the collective lever make the rotor produce more lift

- **A.** The rotational speed of every blade rises, and with it the chord
- **B.** The pitch angle of every blade rises, and with it the angle of attack
- **C.** The plane of rotation of every blade rises, and with it the relative wind

36. The bell cranks from the cockpit controls are traced to the swash plate while the blade control rods run from its other part. Which part do the bell cranks move?

- **A.** Both parts together
- **B.** The rotating part
- **C.** The stationary part

37. Why does the advancing blade flap up

- **A.** Its centre of gravity moves closer to the blade tip
- **B.** It experiences decreased relative wind and generates less lift
- **C.** It experiences increased relative wind and generates more lift

38. Which cockpit control does the pilot use to hold off the drift caused by translating tendency

- **A.** The pedals, which tilt the tail rotor thrust line
- **B.** The collective lever, which lifts the main rotor disk
- **C.** The cyclic stick, which tilts the main rotor disk

EQUIPMENT AND FURNISHINGS

39. Which two items are special features of the integrated cabin entertainment and service system rather than subsystems of it

- **A.** The Cabin Lighting System and the Cabin Interphone System
- **B.** The Cabin Configuration System and the ACESS Test System
- **C.** The Passenger Services System and the Passenger Address System

40. Which cabin functions can the crew and the passengers reach through the NICE system

- **A.** Window-shades, galley power, audio/video and seat heating
- **B.** Illumination, cabin pressure, audio/video and internet access
- **C.** Illumination, window-shades, audio/video and internet access

Answer key

Answer, explanation and the manual page each question comes from

1. **A** The HSI is the instrument that combines compass and radio navigation information. (L2, PDF p. 361)
2. **C** The troposphere lapse rate is 0.65 °C per 100 m of climb. (L2, PDF p. 365)
3. **A** Fluctuating indications are the sign of water lying in the pipes. (L1, PDF p. 443)
4. **C** Electrical indicators have no mechanism to turn, so the correction is done electrically. (L2, PDF p. 459)
5. **C** The indicated airspeed is a dynamic pressure indicator, and at low altitudes the limit depends on the dynamic pressure acting on the aircraft structure, so that limit can be marked as a speed. (L3, PDF p. 465)
6. **B** The airspeed indicator measures with a differential pressure capsule and the altimeter with an aneroid capsule, and a mach meter contains both of those bellows. (L3, PDF p. 473)
7. **B** These are the air data outputs; attitude and position come from the other two sections. (L2, PDF p. 541)
8. **C** It is a caution level alert with Master Caution and an owl tone. (L3, PDF p. 567)
9. **B** The gyroscope makes every effort to maintain the direction of its axis in space. (L2, PDF p. 619)
10. **A** The horizontal component of apparent wander is apparent drift. (L1, PDF p. 627)
11. **B** Real errors from construction are error values over and above those due to the earth's rotation and transport. (L3, PDF p. 633)
12. **B** The vertical axis gyro is the attitude gyro, so it does attitude measurement. (L2, PDF p. 651)
13. **A** With the axis in earth vertical the balls are balanced and no momentum acts on the gyro axis. (L3, PDF p. 663)
14. **A** The roll level switch activates the roll torque motor, whose force precesses the gyro back into the correct direction. (L3, PDF p. 673)
15. **A** A linear spring makes the precession angle proportional to the angular velocity about the measuring axis. (L3, PDF p. 705)
16. **A** In DG mode nothing corrects the drift automatically, so it is set out by hand. (L3, PDF p. 785)
17. **A** The rose is the heading indication; the course bar and the distance are navigation system indications. (L3, PDF p. 795)
18. **B** With the approaching minimums aural the call-out normally comes on 80 feet above the decision height; the 100 feet figure belongs to the plus hundred call-out of the same option. (L2, PDF p. 919)
19. **C** Two envelopes are computed, one for caution level and one for warning level. (L1, PDF p. 927)
20. **B** Once the relays are energised the conflicting ground is painted solid yellow or solid red. (L3, PDF p. 935)
21. **A** In an integral regulator it is how fast the output moves that follows the input. (L1, PDF p. 72)
22. **A** The pilot has to change attitude first and let the path follow from it. (L3, PDF p. 74)
23. **B** Vertical guidance comes from the pitch channel together with the thrust system under AP/FD control. (L3, PDF p. 80)
24. **B** They are the building blocks the outer loops drive in order to steer the flight path. (L3, PDF p. 92)
25. **C** With the standing moment carried by the stabilizer and the elevator at neutral, nothing springs back at disengagement. (L3, PDF p. 110)
26. **A** Each channel takes its basic mode at once, pitch attitude in pitch and roll attitude or heading in roll. (L3, PDF p. 124)
27. **C** The selected heading may be changed at any time while heading select is running. (L2, PDF p. 136)
28. **B** Because the path is built of segments, each one has to be intercepted cleanly and then held. (L3, PDF p. 144)
29. **C** The same controller drives the rate down to nothing and then keeps it there. (L2, PDF p. 162)
30. **A** Because the controller thinks it is higher than it is, it keeps commanding a descent and so forces a positive touch down. (L3, PDF p. 212)
31. **A** Alpha floor protection answers an extreme angle of attack by applying go around thrust. (L3, PDF p. 250)
32. **B** The approval must cover IFR flight before all weather operation can follow. (L2, PDF p. 270)
33. **C** CAT II minima permit a manual landing without requiring one. (L2, PDF p. 280)
34. **A** The blade moving forward is the advancing blade. (L1, PDF p. 10)
35. **B** More pitch means more angle of attack on every blade, so the rotor makes more lift. (L3, PDF p. 40)

36. **C** Cockpit inputs arrive through the bell cranks at the stationary part, and the rotating part passes the movement on to the blades. (L3, PDF p. 44)

37. **C** More relative wind means more lift, so it flaps up. (L2, PDF p. 50)

38. **C** Tilting the rotor disk is a cyclic input, so the cyclic stick is the control that holds off the drift. (L3, PDF p. 56)

39. **B** The CCS and the ATS are named as the two special features added to the five subsystems. (L3, PDF p. 336)

40. **C** Illumination, window-shades, audio/video and internet access are the cabin functions printed. (L2, PDF p. 338)